

MASTER PROMPT FOR CLAUDE

You are the Lead Software Architect, Principal AI Engineer, Senior Backend Engineer, Senior Frontend Engineer, Senior DevOps Engineer, Senior Database Architect, Senior BIM/CAD Engineer and Technical Writer for this project.

You have already received the complete master specification of the AI Home Designer platform.

Your task is NOT to redesign or change the architecture.

Your task is to transform the existing specification into a 100% implementation-ready technical documentation set.

Rules

- Do not simplify anything.
- Do not replace existing decisions.
- Preserve the current architecture.
- Expand every module to implementation level.
- Use deterministic outputs.
- Every document must be internally consistent.
- Assume this documentation will be used by multiple engineers simultaneously.

The goal is that a new engineering team can build the entire platform without making architectural decisions.

Generate ALL remaining implementation documents.

Include every missing detail.

At minimum produce the following documents.

1. Complete PostgreSQL Documentation

Include:

- Full CREATE TABLE statements
- Constraints
- Foreign keys
- Primary keys
- Composite indexes
- Unique indexes
- Partial indexes

- Triggers
- Functions
- Views
- Materialized views
- Audit tables
- Soft delete strategy
- Versioning strategy
- Migration strategy
- Seed data
- Performance recommendations

2. Complete OpenAPI 3.1 Specification

Generate:

- Full OpenAPI YAML
- Every endpoint
- Every request
- Every response
- Every DTO
- Every error model
- Authentication
- Authorization
- Pagination
- Filtering
- Sorting
- Examples
- Webhooks

No placeholders.

3. Backend Implementation Specification

For every module generate:

Folder structure

Controllers

Services

Repositories

DTOs

Entities

Validators

Middlewares

Guards

Interceptors

Filters

Dependency graph

Business logic

Error handling

Caching strategy

Transactions

Logging

Metrics

Background jobs

Configuration

Testing strategy

4. Frontend Implementation Specification

Describe every React component.

For every component include:

Purpose

Props

State

Events

Hooks

API calls

Validation

Loading states

Error states

Responsive behavior

Accessibility

Dependencies

Folder structure

5. Canvas Engine Specification

Describe completely:

Drawing engine

Geometry engine

Snapping

Selection

Dragging

Resize

Rotation

Boolean operations

Measurements

Undo/Redo

History

Infinite canvas

Coordinate system

Grid

Performance optimization

Object serialization

6. 3D Engine Implementation

Include:

Scene graph

Three.js architecture

Mesh generation

Materials

Textures

Lighting

Camera

LOD

Raycasting

Collision detection

Navigation

Animations

Memory optimization

GPU optimization

7. AI Orchestration Implementation

Specify:

Every AI agent

Inputs

Outputs

Prompt routing

Memory

Retries

Fallbacks

Validation

Agent communication

Execution graph

Caching

Token optimization

Model selection

Failure recovery

Monitoring

8. Geometry Algorithms

Describe algorithms for:

Room generation

Wall generation

Door placement

Window placement

Roof generation

Stair generation

Collision detection

Constraint solving

Optimization

Polygon validation

Coordinate precision

9. Cost Engine

Include:

Material quantity calculation

Labor estimation

Regional pricing

Supplier integration model

Waste calculation

Tax calculation

Profit calculation

Budget comparison

Cash flow estimation

10. Energy Simulation Engine

Specify:

Thermal model

Heat loss

Solar gain

Ventilation

Insulation

HVAC estimation

Energy class

Annual consumption

CO₂ estimation

11. BIM Engine

Describe:

IFC mapping

Entity mapping

Property sets

Export rules

Import rules

Validation

BCF support

Future Revit compatibility

12. Import/Export Engines

Generate complete implementation documentation for:

PDF

DXF

IFC

OBJ

GLTF

SVG

PNG

JSON

Include parsing algorithms and validation.

13. Security Documentation

Include:

Threat model

OWASP mitigation

Authentication flow

Authorization

Secrets management

Encryption

Key rotation

Rate limiting

Audit logging

Compliance

GDPR considerations

14. DevOps Documentation

Generate:

Dockerfiles

Docker Compose

Kubernetes manifests

Helm charts

Terraform layout

CI/CD pipelines

GitHub Actions

Monitoring

Prometheus

Grafana

Alerting

Logging

Backup

Disaster recovery

15. Testing Documentation

Generate:

Unit tests

Integration tests

E2E tests

Visual regression tests

Load tests

Performance benchmarks

Security tests

AI regression tests

Target >90% coverage.

16. Developer Documentation

Generate:

Coding standards

Naming conventions

Architecture decision records

Contribution guide

Branch strategy

Release process

Versioning

Documentation standards

17. UX Documentation

Include:

User journeys

Wireframes (text)

Interaction rules

Animations

Micro-interactions

Keyboard shortcuts

Accessibility

Responsive behavior

18. Plugin SDK

Design a complete extension system.

Include:

Plugin lifecycle

Events

Permissions

API

Sandbox

Versioning

Marketplace

19. Mobile Implementation

Specify:

React Native architecture

Offline synchronization

Caching

Notifications

Authentication

Viewer

Basic editing

20. Production Runbook

Generate:

Deployment checklist

Monitoring checklist

Incident response

Rollback

Scaling

Maintenance

Upgrade strategy

Recovery procedures

Output Rules

Never summarize.

Never skip details.

Never use placeholders.

Never answer "implementation depends".

Always provide production-grade documentation.

Continue automatically from one document to the next until every missing implementation document has been completed.

Treat the existing master specification as immutable and expand it into a complete engineering handbook suitable for immediate software development.

MASTER PROMPT – AUTOMATED TECHNICAL DOCUMENTATION & PERMIT PACKAGE

You are continuing the existing AI Home Designer Master Specification.

Do NOT redesign the platform.

Do NOT change any previous architecture.

Your task is to expand the specification by adding a complete Technical Documentation & Permit Documentation Engine.

This module must become a core part of the platform.

The generated documentation must be detailed enough that a software engineering team can implement it directly.

OBJECTIVE

The platform must be capable of automatically generating nearly all architectural documentation required for residential buildings.

The system must support the Romanian market first while being designed to support additional countries in the future through country-specific regulation packs.

The platform must clearly distinguish between:

- Automatically generated documentation
- Documentation requiring architect review
- Documentation requiring licensed engineer approval
- Documents that require signatures and stamps according to local law

Never assume AI can legally replace licensed professionals.

Instead, AI prepares the complete documentation package for professional review.

CREATE A COMPLETE MODULE

Generate the following implementation documents.

1. Technical Documentation Engine

Design the complete architecture.

Include:

Document generation pipeline

Template engine

Drawing generator

Annotation engine

Dimension engine

Revision system

Document validation

Versioning

Metadata

Export engine

Digital signatures support

2. Drawing Generator

Automatically generate

Site Plan

Floor Plans

Roof Plan

Foundation Plan

Sections

Elevations

Door Schedule

Window Schedule

Room Schedule

Furniture Plan

Dimensioned Drawings

Structural Grid

Legend

North Arrow

Scale

Title Block

Revision Table

3. Technical Report Generator

Automatically generate

Project description

Building description

Construction system

Materials

Structural assumptions

Fire safety summary

Accessibility summary

HVAC summary

Electrical summary

Water supply summary

Drainage summary

Thermal insulation summary

Construction methodology

Maintenance recommendations

4. Material Documentation

Generate

Bill of Quantities (BOQ)

Material Schedule

Material Quantities

Supplier-ready Purchase Lists

Waste Estimation

Cost Breakdown

Transportation Estimate

Storage Recommendations

5. Construction Documentation

Automatically create

Construction Phases

Construction Timeline

Inspection Checklist

Quality Checklist

Daily Work Schedule

Material Delivery Schedule

Equipment Schedule

Risk Assessment

Critical Path

6. Romanian Permit Documentation

Design a dedicated Romania module.

Include support for

DTAC

PTh

DDE

Urban planning documentation

Building statistics

Area calculations

Compliance reports

Permit package assembly

Explain which documents can be fully generated automatically and which require licensed architect or engineer approval.

7. Country Regulation Packs

Design an extensible regulation engine.

Support

Romania

Hungary

Germany

France

Italy

Spain

UK

USA

Each regulation pack must define

Minimum room sizes

Accessibility

Fire regulations

Energy requirements

Parking

Setbacks

Roof rules

Building height

Occupancy

Validation rules

8. Architect Review Mode

Implement a workflow where

AI generates documentation

↓

Architect reviews

↓

Architect edits

↓

Architect approves

↓

Final documentation exported

Support

Revision history

Comments

Approval status

Digital signatures

Audit trail

9. Professional Report Generator

Automatically generate

Client Report

Architect Report

Engineer Report

Construction Report

Energy Report

Cost Report

Permit Summary

Risk Report

10. PDF Generator

Generate professional PDFs including

Company logo

Project metadata

Automatic page numbering

Revision table

Headers

Footers

Table of contents

Cross references

Bookmarks

QR code

Digital signature area

11. Compliance Engine

Automatically verify

Romanian building regulations

Accessibility

Fire safety

Room sizes

Energy requirements

Documentation completeness

Drawing consistency

Missing information

Generate a compliance report.

12. Legal Workflow

Clearly distinguish

AI-generated content

Professional-reviewed content

Professionally approved content

Signed documentation

Permit-ready documentation

Never state that AI replaces licensed architects or engineers.

13. Future Expansion

Design support for

Electronic permit submission

Government APIs

Digital signatures

BIM approval workflow

Municipality integrations

Inspection tracking

Construction monitoring

Digital Twin updates

OUTPUT RULES

Never summarize.

Never skip implementation details.

Never use placeholders.

Generate production-grade technical specifications.

Continue automatically until the entire Technical Documentation & Permit Documentation module is fully specified.

The final documentation must integrate seamlessly with the existing AI Home Designer architecture and elevate the platform into a complete AI-assisted architectural documentation system suitable for professional use.